Test report 5 – STEP export and kernel comparison

2026-10-23 05:04 · Commit ff21522 (worktree-agent-a8e28dbf90111aa26)

46/46	6.8e-06	0	101/855
Examples: STEP valid, closed, volume/area/bounding box as in the kernel	largest rel. volume deviation OCCT ↔ SMLib	approximated faces/edges (everything written exactly)	nodes equal with solidcore (rest unsupported/blocked)

1. Scope

- **Facade** (`smlib/facade`, ABI version 2 → 3): the new query `sc_brep(h)` returns topology (bodies → shells → faces → loops → oriented edges, vertices) and exact geometry: every face and edge as NURBS (degree, knots, weights, control points), analytic surfaces (plane, cylinder, cone, sphere, torus, surface of revolution, extrusion surface) additionally with their parameters. Non-B-spline geometry (blends, offsets) would be approximated to the model tolerance and flagged. Native side (FFI) and Wasm bridge (`smlib-bridge.js`, `exports.txt`).
- **Neutral B-rep** `cad_kernel::brep` with `Kernel::brep` (default: Unsupported). `kernel-smlib` checks the analytic parameters against the NURBS (9×9 samples), trims edge curves to their interval (knot insertion), detects exact lines and circles and determines `same_sense` from SMLib's normal at an interior face point.
- **Writer** `cad-step` (kernel-independent): AP242 (default) or AP214, product structure (`PRODUCT` … `SHAPE_DEFINITION_REPRESENTATION`), units mm and degrees, `MANIFOLD_SOLID_BREP` / `BREP_WITH_VOIDS` per body (several shells → several solids), open surfaces as `SHELL_BASED_SURFACE_MODEL`, colors from `color()` as `STYLED_ITEM` per face.
- **Integration**: `Engine::export_step`, `cad export -o x.step [--schema ap214]` (JSON with volume, area, bounding box and statistics), session `Export{format:"step"}` (additive, `protocol.ts` regenerated), Wasm.
- **Verification** `tests/step/verify.py`: reads every file with OpenCASCADE (not our code), `BRepCheck_Analyzer`, closed shells, number of solids/faces, volume and bounding box against the kernel (relative 0.0001), area (0.0005, see section 7). Rust tests of the writer (syntax, references, numbers read back with our own Part 21 reader) run without Python.
- **solidcore**: crate `kernel-solidcore`, CLI `cad diff-kernels`, documentation `docs/solidcore-umstieg.md`.

Why AP242: AP242 is the current edition that merges and replaces AP203 and AP214; the B-rep entities (Part 42) are the same in both, the choice only changes the header and the application context. AP214 remains available via `--schema ap214` for older readers that do not know the AP242 schema name.

2. Environment

Date	2026-10-23 05:04
Commit	ff21522 (worktree-agent-a8e28dbf90111aa26)
Operating system	Linux 6.18.44-fc-v153, 4 cores
Rust	rustc 1.97.0 (2d8144b78 2026-07-07)
C++	g++ (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
SMLib	tangency-labs/smlib capi-tess @ 7eba8aa, facade version 3 (sc_brep)
solidcore	tangency-labs/solidcore main @ 3a6a50b (git dependency, feature `solidcore`)
STEP reader	OpenCASCADE via cadquery-ocp 8.0.1.1.0 (Python 3.13.16, venv /opt/tools/venv-step)
Node	v22.22.0

3. Check results

`cargo test --workspace`	ok	243 passed, 0 failed
`cargo clippy --workspace --all-targets`	ok	1 warning in crates/cad-eval/src/edit.rs
`node crates/kernel-smlib/js/test-node.mjs`	ok	11 tests, all node tests passed
`tests/openscad-compat/compare.py`	ok	33/33 files match
`tests/step/verify.py`	ok	46/46 passed
`cad diff-kernels` **(all examples)**	ok	Exit 1 (1 = not all nodes equal)

4. All examples: kernel ↔ STEP file read back

V: volume in mm³ (SMLib `mass_props` or OCCT, adaptive integration 1e-9). dV, dA: relative deviation of volume and area; dBox: largest deviation of a bounding box coordinate relative to the diagonal. "valid": BRepCheck valid, all shells closed, number of solids and faces as written. Face types: as written (OCCT reads the same types).

Model	Solids	Faces	V kernel	V STEP (OCCT)	dV	dA	dBox	valid	Face types	Curve types
01_cube_variants	5	30	3208.000	3208.000	1.4e-16	0.0e+00	0.0e+00	yes	plane 30	line 60
02_sphere_variants	4	4	1017.876	1017.876	7.0e-12	5.4e-05	1.9e-17	yes	sphere 4	circle 4
03_cylinder_variants	5	15	2104.867	2104.867	4.7e-09	1.1e-05	7.1e-17	yes	cylinder 5, plane 10	circle 10, line 5
04_cone_variants	4	11	1342.507	1342.507	1.5e-08	6.5e-06	3.6e-17	yes	cone 4, plane 7	circle 7, line 4
05_nested_booleans	2	25	8312.337	8312.281	6.8e-06	3.2e-05	1.4e-09	yes	cylinder 7, plane 17, sphere 1	circle 19, line 31, nurbs 17
06_hole_grid	1	30	6295.221	6295.221	5.8e-09	2.5e-05	0.0e+00	yes	cylinder 24, plane 6	circle 48, line 36
07_crown_ring	1	67	2160.960	2160.960	3.0e-09	4.1e-05	1.4e-16	yes	cylinder 17, plane 50	circle 34, line 161
08_module_children	2	28	6172.832	6172.832	1.2e-09	2.6e-06	1.1e-17	yes	cylinder 8, plane 20	circle 16, line 32
09_rotate_mirror	7	39	1460.097	1460.097	1.6e-08	1.3e-07	0.0e+00	yes	cylinder 1, plane 38	circle 2, line 73
10_scale_multmatrix	4	24	3400.000	3400.000	2.7e-16	2.0e-16	1.0e-16	yes	bspline 4, plane 20	line 48
11_linear_extrude_basic	3	17	1513.000	1513.000	1.5e-16	0.0e+00	0.0e+00	yes	plane 17	line 33
12_linear_extrude_twist	3	18	2660.002	2660.002	4.7e-08	6.2e-07	1.8e-09	yes	bspline 12, plane 6	bspline 8, line 28
13_rotate_extrude_ring	2	5	1307.931	1307.931	4.4e-08	7.0e-05	1.8e-09	yes	plane 2, revolution 2, torus 1	circle 6, line 2
14_pipe_bend	1	8	1574.897	1574.897	3.2e-08	7.3e-05	2.1e-09	yes	cylinder 4, plane 2, revolution 2	circle 12, line 4
15_vase	1	11	6364.867	6364.865	2.3e-07	2.0e-05	1.7e-09	yes	plane 3, revolution 8	circle 10, line 8
16_polygon_holes	2	23	2318.000	2318.000	5.9e-16	1.6e-16	0.0e+00	yes	plane 23	line 57
17_polyhedron	2	10	626.667	626.667	1.8e-16	0.0e+00	4.8e-17	yes	plane 10	line 17
18_functions_let	1	16	7040.000	7040.000	2.6e-16	3.0e-16	0.0e+00	yes	plane 16	line 42
19_hex_nut	2	16	264.623	264.623	4.6e-09	1.6e-05	3.3e-17	yes	cylinder 2, plane 14	circle 4, line 32
20_box_with_lid	2	70	20575.184	20575.184	7.5e-10	6.9e-07	0.0e+00	yes	cylinder 19, plane 51	circle 54, line 119
21_angle_bracket	1	23	10407.841	10407.841	6.4e-10	4.8e-06	0.0e+00	yes	cylinder 8, plane 15	circle 16, line 47
22_gear	1	87	6943.665	6943.665	2.3e-09	7.1e-06	1.4e-17	yes	cylinder 5, plane 82	circle 10, line 245
23_shelf_holder	1	16	10423.680	10423.680	2.9e-09	5.3e-07	1.5e-09	yes	cone 1, cylinder 4, plane 11	circle 9, line 27, nurbs 2
24_modifiers	2	13	3622.345	3622.345	1.4e-09	6.9e-06	0.0e+00	yes	cylinder 1, plane 12	circle 2, line 25
25_root_modifier	1	8	346.608	346.608	7.1e-09	1.0e-05	5.6e-17	yes	cone 1, plane 7	circle 2, line 13
26_if_else_variants	4	18	1662.060	1662.060	2.4e-09	1.4e-05	1.6e-17	yes	cylinder 3, plane 14, sphere 1	circle 7, line 27

27_spacers_2d	5	24	2962.093	2962.093	2.0e-09	1.7e-05	1.7e-16	yes	cylinder 9, plane 15	circle 18, line 24
28_cable_clip	2	15	561.074	561.074	9.0e-09	5.7e-05	1.8e-16	yes	cylinder 5, plane 10	circle 10, line 23
29_assert_strings	1	8	1907.434	1907.434	5.6e-11	2.8e-06	0.0e+00	yes	cylinder 2, plane 6	circle 4, line 14
30_each_ranges	1	32	832.000	832.000	4.1e-16	1.8e-16	0.0e+00	yes	plane 32	line 90
31_phone_stand	1	18	40840.229	40840.229	8.9e-16	0.0e+00	1.2e-16	yes	plane 18	line 60
32_low_fn_shapes	5	41	1199.037	1199.037	1.9e-16	9.5e-16	2.9e-17	yes	plane 41	line 92
33_global_fn	2	6	1629.439	1629.439	7.1e-09	6.2e-05	9.3e-17	yes	cylinder 2, plane 3, sphere 1	circle 5, line 2
dose_shell	2	52	54459.131	54459.131	4.0e-10	1.4e-06	2.1e-16	yes	cylinder 24, plane 12, sphere 16	circle 48, line 48
fase_chamfer	4	43	13997.770	13997.771	1.4e-08	1.2e-06	1.3e-09	yes	cylinder 1, plane 40, revolution 2	circle 6, line 93
fillet_zonen	2	27	17123.696	17123.696	7.6e-09	1.8e-06	1.7e-09	yes	cylinder 11, plane 12, revolution 2, sphere 2	bspline 4, circle 18, line 35
flasche	2	19	164188.200	164188.120	4.8e-07	2.6e-05	9.4e-10	yes	cone 1, cylinder 4, plane 6, revolution 8	bspline 2, circle 19, line 16
freiformdeckel	2	32	6280.193	6280.196	4.9e-07	1.9e-04	1.3e-09	yes	bspline 6, cylinder 12, plane 6, sphere 8	bspline 8, circle 24, line 28
gehaeuse	2	20	8860.000	8860.000	2.1e-16	2.6e-16	0.0e+00	yes	plane 20	line 47
griffschale	1	49	11646.797	11646.797	3.4e-09	6.2e-06	0.0e+00	yes	cylinder 4, plane 43, revolution 2	circle 12, line 112
kabelfuehrung	2	36	7748.147	7748.147	7.7e-10	8.3e-05	7.2e-10	yes	bspline 8, plane 28	bspline 13, line 77
knauf_mirror	2	10	15681.708	15681.686	1.4e-06	2.2e-05	8.8e-10	yes	bspline 2, cylinder 3, plane 3, revolution 2	bspline 8, circle 4, line 3
lochplatte	2	23	2376.681	2376.681	4.9e-08	1.1e-05	1.7e-09	yes	cylinder 5, plane 10, revolution 8	circle 26, line 23
loft_sweep	3	17	6529.142	6529.145	5.5e-07	7.1e-06	9.9e-10	yes	nurbs 5, plane 12	bspline 5, circle 10, line 18
lueftungsgitter	1	98	9236.448	9236.448	7.4e-14	9.3e-14	2.1e-18	yes	plane 98	line 232
winkel_anker	3	34	8337.812	8337.812	1.5e-09	5.7e-06	1.7e-17	yes	cylinder 6, plane 28	circle 12, line 54

Largest deviations: volume 6.8e-06, area 1.9e-04, bounding box 2.1e-09. The volume deviations above 1e-7 concern B-spline surfaces and surfaces of revolution, where both sides integrate numerically.

Totals over all examples

Face type (STEP entity)	Count	Curve type	Count
plane PLANE	926	line LINE	2267
cylinder CYLINDRICAL_SURFACE	196	circle CIRCLE	488
revolution SURFACE_OF_REVOLUTION	36	bspline B_SPLINE_CURVE_WITH_KNOTS	48
sphere SPHERICAL_SURFACE	33	nurbs … + RATIONAL_B_SPLINE_CURVE	19

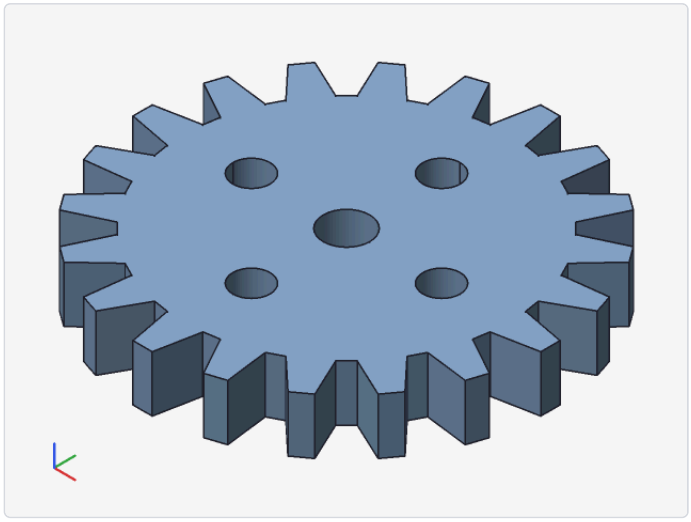
bspline B_SPLINE_SURFACE_WITH_KNOTS	32
cone CONICAL_SURFACE	7
nurbs … + RATIONAL_B_SPLINE_SURFACE	5
torus TOROIDAL_SURFACE	1

5. STEP geometry read back

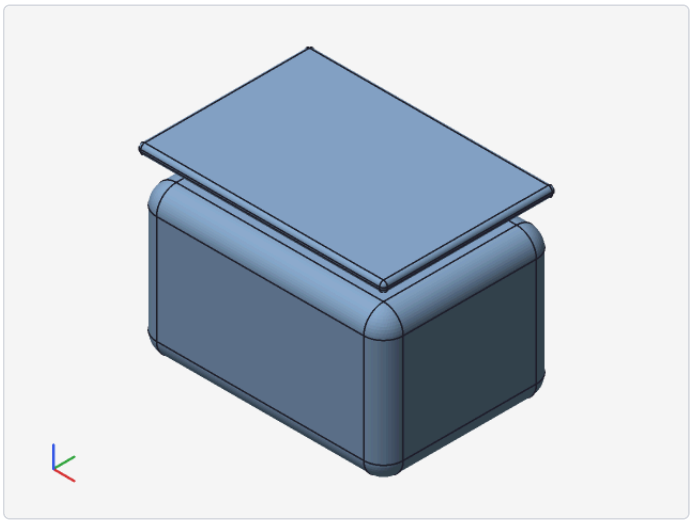
OpenCASCADE reads the STEP file, meshes it (chord error 0.2 % of the diagonal) and writes STL; the edge lines are the B-rep edges of the shape read in. Rendered with the CLI's software rasterizer (`cad-render`, example `render_stl`). The images therefore show the geometry as seen by a different kernel.



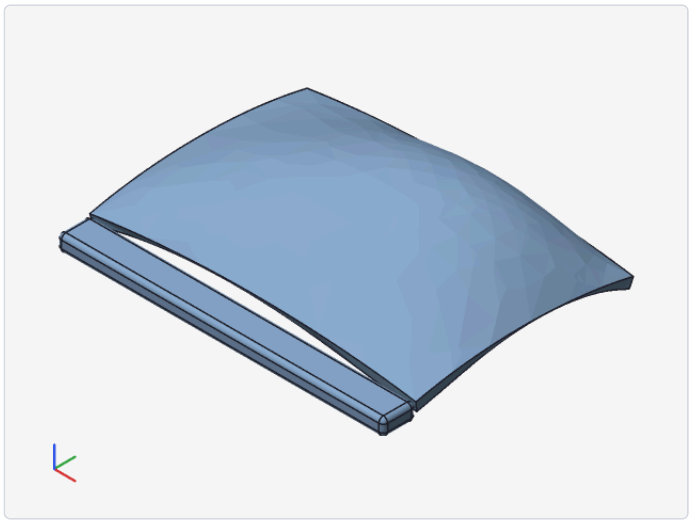
14_pipe_bend – pipe bend: surfaces of revolution and cylinders



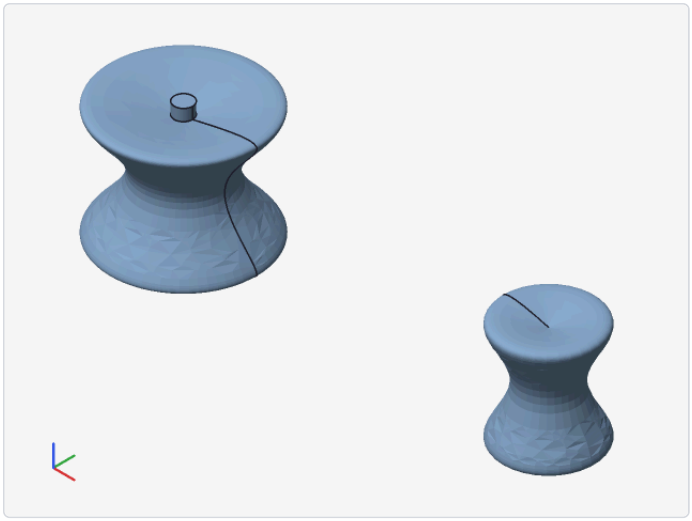
22_gear – gear: 82 planes, 5 cylinders



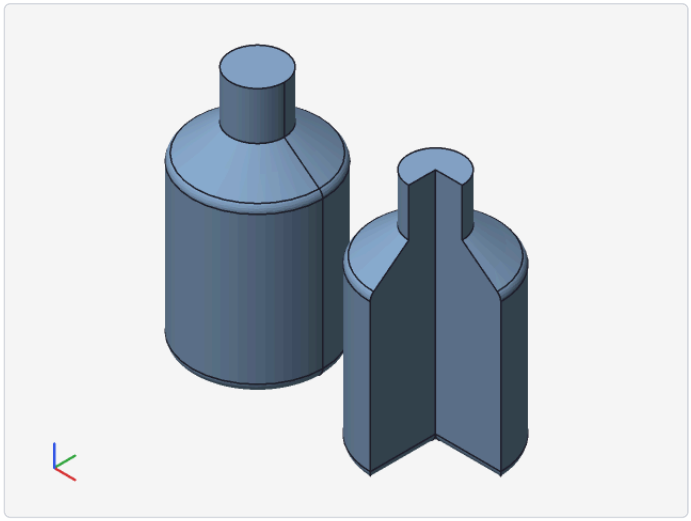
dose_shell – can (shell, fillets): cylinders and spheres



freiformdeckel – freeform lid: B-spline surfaces



knauf_mirror – knob (mirrored): revolution, B-spline



flasche – bottle: surfaces of revolution, cone

6. Kernel comparison `cad diff-kernels` (SMLib ↔ solidcore)

solidcore @ `3a6a50b`, all 46 examples, 855 evaluated nodes: **101** equal, **267** unsupported, **487** blocked (input unsupported), **0** different, **0** failed. Only 2D nodes are equal (profiles are computed in Rust without the kernel). Every primitive aborts in solidcore with `Unported: SmSurface::DropCurve (intersect.surface_drop_lib_p3)`; booleans, queries and mass properties can therefore not be checked yet. The comparison runs over all files without crashing.

Operation (node)	Nodes	equal	unsupported	blocked	different
`box`	94	0	93	1	0
`sphere`	8	0	8	0	0
`cylinder`	110	0	110	0	0
`torus`	2	0	2	0	0
`polyhedron`	8	0	8	0	0
`linear_extrude`	32	0	32	0	0
`rotate_extrude`	8	0	8	0	0
`loft`	1	0	1	0	0
`sweep`	4	0	4	0	0
`surface`	1	0	1	0	0
`difference`	42	8	0	34	0
`union`	9	1	0	8	0
`intersection`	2	0	0	2	0
`translate`	255	21	0	234	0
`rotate`	46	0	0	46	0
`fillet`	5	0	0	5	0
`chamfer`	1	0	0	1	0
`shell`	2	0	0	2	0
`zone`	6	0	0	6	0
`circle`	23	23	0	0	0
`group`	99	2	0	97	0
`polygon`	13	13	0	0	0
`rect`	22	22	0	0	0

Further labels (module calls, groups, 2D primitives) in the JSON `target/step/diff-kernels.json`. Acceptance and missing parts: `docs/solidcore-umstieg.md`.

7. Known limitations

- **Area:** SMLib integrates areas only to about 1e-4 (`SmApiBrepComputeArea` clamps the accuracy to [1e-4, 1e-1]). For `freiformdeckel` the kernel area deviates by 1.9e-4; OCCT adaptive (6658.641 mm²) and a fine OCCT mesh (6658.656 mm²) agree with each other, so the error lies in the kernel reference value. The area check therefore uses 5e-4, volume and bounding box 1e-4.
- OCCT's default integration (fixed Gauss rule) deviates by up to 1e-4 for B-spline surfaces and surfaces of revolution; verify.py integrates adaptively (1e-9).
- No p-curves (2D curves in the surface parameter space): readers project the 3D edges. OCCT does this without errors for all examples; other systems might read B-spline surfaces with a seam more slowly.
- Edge curves of booleans are B-splines (intersection curves); exact circles/lines are detected (tolerance 1e-9 of the model size), ellipses only if SMLib holds them as such (none in the examples).
- Surfaces without a B-spline form (blend, offset surfaces) would be approximated; in all examples SMLib supplies exact B-splines or analytic surfaces (0 approximated).
- Voids (`BREP_WITH_VOIDS`) are checked in the Rust test and on the kernel (sphere minus sphere); none occur in the examples.
- Colors: per face from `color()` as `STYLED_ITEM` ; OCCT reads them only via XCAF, verify.py does not check them.
- The root is exported in its own coordinate system (like STL/3MF); 2D results are rejected.
- In the browser the session writes a fixed timestamp (`1970-01-01T00:00:00`), the CLI the current UTC time.
- solidcore: see section 6; `kernel-solidcore` only covers the API entry points available today.